Peer Review

Report Title: A Low-Cost and Low-Complexity Motion Sensor System Using IMU and Video-Based Tracking

Report Authors: Jack Hutton, Alexander Jonsson, Sahand Khodabandehloo

Comments on the report:

Was the objective of the report easy to understand?

Yes, the objective of the report is very clearly formulated. The authors explain the motivation behind the project well, especially the need to bridge the gap between laboratory-grade biomechanical validation and practical tools that can be used in everyday training environments. The research question is clearly stated, and the aims of developing the browser based MoveSync application are concrete and well defined. The system requirements are also listed clearly, which makes it easy to understand the intended scope of the project.

How is the background described? Is there an introduction and overview of the field?

The background section is comprehensive and well written. It provides a solid overview of motion capture techniques, IMU sensors, sensor fusion methods, and video-based pose estimation. The explanation of the differences between PoseNet and MoveNet is particularly clear, and the justification for selecting MoveNet Thunder for offline video processing is convincing.

Overall, the background demonstrates good understanding of the theoretical foundations of the system. The connections between sports performance analysis and the chosen technologies are also explained clearly.

Is the method well described?

Most parts of the methodology are well structured and logically presented. The modular client server architecture and the design of the web application are clearly explained. The description of the session based data model and client side execution environment shows thoughtful system design.

The choice of the Madgwick algorithm for sensor fusion is clearly documented, including the fixed gain parameter $\beta = 0.1$. The mathematical explanation of computing relative rotation quaternions and extracting joint angles using

$$\theta=2\arccos(w_rel)$$

is presented properly and demonstrates a solid theoretical basis.

However, one clear area for improvement is the signal processing section. The authors state that the low-pass (10 Hz) and high-pass (0.5 Hz) filter cut-off frequencies were chosen "arbitrarily through trial and error." This weakens the academic strength of the report. It would be interesting to know whether the group plans to validate the joint angles against a reference system.

Are the results presented clearly?

For a progress report, the results are presented clearly and honestly. The figures comparing unfiltered and filtered acceleration data effectively demonstrate how bias and noise are reduced. This clearly shows that the filtering stage is functioning as intended.

The authors also openly acknowledge that the computed elbow and wrist angle time series currently contain "significant errors, with physically impossible angles." This transparency is a strong point, especially in a progress report context.

However, it would have improved the results section if the authors briefly discussed possible reasons for these incorrect angle values. For example, issues related to calibration, drift, quaternion alignment, or synchronization errors could be considered. Including a short hypothesis would demonstrate deeper troubleshooting and critical reflection.

Were there any parts that were difficult to understand?

The text itself is generally very clear and easy to follow. The explanations of both the system architecture and the mathematical concepts are understandable.

The only unclear part concerns why the joint angle calculations are failing so significantly. While the authors mention that errors exist, the lack of preliminary analysis of the cause makes this section feel somewhat incomplete. A short discussion of potential error sources would strengthen the reader's understanding.

Strong Points of the Report

- **Clear purpose and defined scope:** The objectives and system requirements are well structured and realistic.
- **Strong system architecture:** The modular browser-based implementation is well designed and technically ambitious.
- **Good theoretical foundation:** The explanations of IMUs, sensor fusion, and pose estimation are strong and relevant.
- **Transparency:** The authors openly acknowledge current technical limitations instead of presenting incomplete results as final outcomes.

Weak Points of the Report

- **Methodological justification:** The filter cut-off frequencies lack scientific justification and reduce academic rigor.
- **Technical limitation:** The core feature of producing realistic joint angles is currently not functioning correctly.
- **Limited error analysis:** There is little discussion of why the joint angle calculations are producing physically impossible values.
- **Incomplete sections:** The Discussion and Conclusion sections are not yet fully developed.

Overall Assessment

This is an ambitious and well-designed project with significant potential. The foundational theoretical work and system architecture are strong, and the implementation of a fully client-side browser-based motion analysis system is impressive.

As a progress report, it clearly demonstrates substantial development and a good understanding of both the theoretical and practical aspects of the system. The authors show awareness of their current limitations, which is appropriate at this stage. To successfully complete the report, the group should prioritize:

- Debugging and validating the quaternion-based joint angle calculations
- Providing scientific justification for signal processing parameters

Overall, this progress report reflects a strong technical foundation and thoughtful design, with clear next steps needed to reach full scientific validation.